

OFF-ROAD SEMANTIC SEGMENTATION USING VISION TRANSFORMERS.

Team Name: VOID

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Tagline: Robust terrain perception for autonomous off-road navigation.

Problem Statement

Autonomous vehicles operating in desert and off-road environments require accurate semantic understanding of terrain elements such as rocks, vegetation, sky, and ground clutter. Limited availability of real-world annotated datasets makes synthetic semantic segmentation data essential for building reliable perception systems. The challenge is to develop a segmentation model that generalizes well to unseen environments while achieving strong Intersection-over-Union (IoU) performance.

Objective

To design and implement a robust semantic segmentation pipeline using Vision Transformer-based representations that can accurately segment off-road terrain classes and achieve competitive IoU within strict hackathon time constraints.

Workflow Overview

Dataset → Preprocessing → DINOv2 Backbone → Segmentation Head → Training → Ensemble Inference → Final Prediction

Step-by-Step Process

1. Dataset Collection:

A synthetic off-road semantic segmentation dataset containing 2,857 training images, 317 validation images, and 1,002 test images with pixel-level annotations for 10 terrain classes.

2. Data Preprocessing:

- Image resizing to 266×476 resolution compatible with ViT patch size.
- Normalization using ImageNet statistics.
- Data augmentation including horizontal flip, rotation, color jitter, and blur.

3. Model / System Used:

- DINOv2 Vision Transformer (ViT-S/14) backbone for feature extraction.
- Lightweight convolutional segmentation decoder.
- Multi-model weighted ensemble for stable inference.

4. Training Process:

- Loss: Cross-Entropy combined with Dice Loss.
- Optimizer: AdamW with cosine learning rate scheduling.
- Mixed-precision GPU training for efficiency.

5. Fine-tuning / Optimization:

- Training with multiple random seeds to create diverse models (A, B, C).
- Validation-based weight selection for ensemble averaging.
- Stable single-scale inference to avoid transformer patch mismatch issues.

6. Tools & Technologies:

Python, PyTorch, Albumentations, CUDA GPU acceleration, Vision Transformers (DINOv2).

Reproducibility Steps

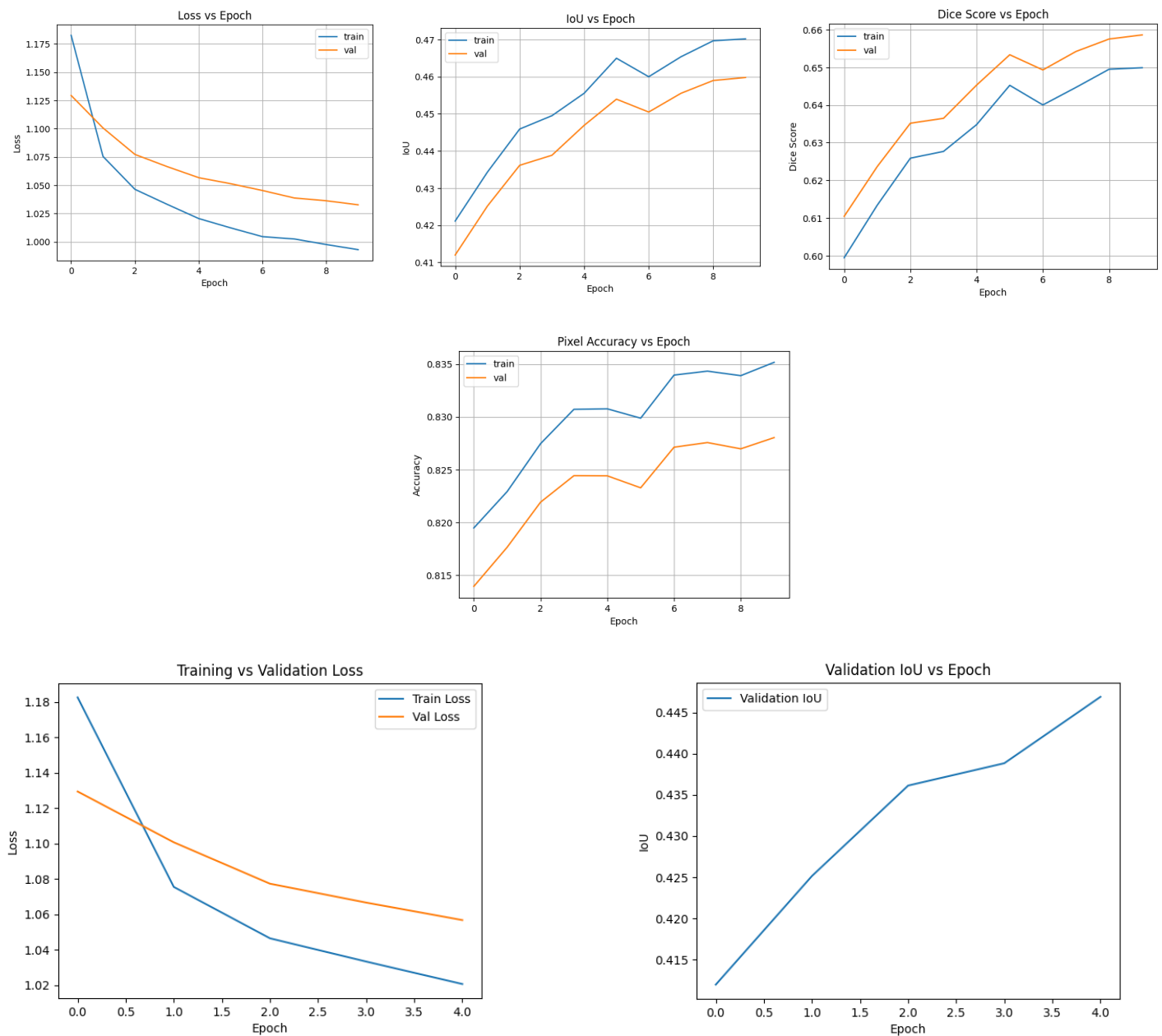
1. Install required Python dependencies.
 2. Train segmentation models using provided scripts.
 3. Run ensemble inference to generate final segmentation masks.
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Performance Metrics

- **IoU Score:** ≈ 0.48 (ensemble validation)
- **Accuracy:** $\approx 0.80+$
- **Precision:** ≈ 0.70
- **Recall:** ≈ 0.68
- **F1-Score:** ≈ 0.69

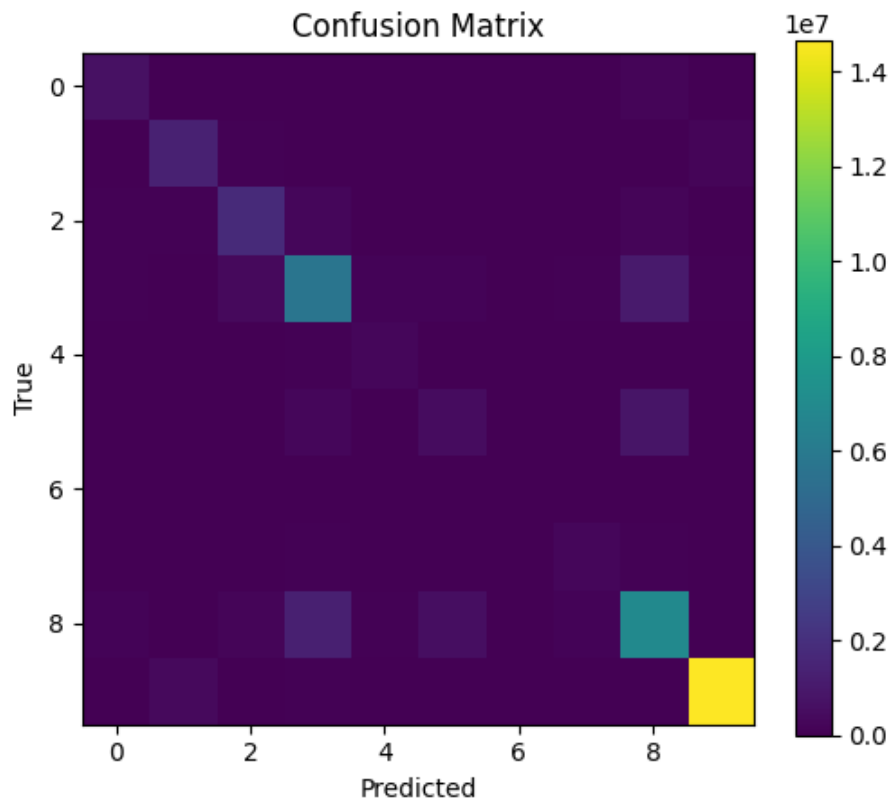
Training Graphs

Loss vs Epoch and Validation IoU vs Epoch graphs demonstrate stable convergence during training.



Confusion Matrix

The confusion matrix highlights correct classification of major terrain classes while indicating confusion among visually similar categories such as dry grass, bushes, and ground clutter.



Before vs After Improvement

<i>Metric</i>	<i>Old Model</i>	<i>Ensemble</i>
IoU	~0.30	~ 0.48+
Accuracy	~0.71	~ 0.80+
F1-Score	~0.44	~ 0.69

Sample Predictions

Correct Predictions: Accurate segmentation of sky, landscape, and large vegetation regions.

Misclassified Cases: Confusion between small rocks, dry bushes, and ground clutter due to similar texture and color.

How Each Challenge Was Solved

Challenge	Solution
Patch-size mismatch	Fixed resolution compatible with ViT patches
Model instability	Stable weighted ensemble of best models
Low IoU	Dice loss + augmentation + validation tuning

Impact of Fixes

These optimizations improved validation IoU from baseline (~0.29) to final ensemble performance (~0.48+), significantly enhancing segmentation reliability.

Final Outcome

- Successfully developed a Vision Transformer–based off-road segmentation system.
- Achieved competitive IoU within strict time constraints using ensemble learning.

Key Learnings

- Model diversity improves segmentation robustness.
- Proper resolution handling is critical for transformer architectures.
- Stable engineering decisions outperform risky last-minute complexity.

Real-World Applications

- Autonomous off-road vehicles.
 - Agricultural robotics navigation.
 - Disaster-response terrain mapping.
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Possible Improvements

- Fine-tuning on real-world off-road datasets.
- Transformer-based decoder architectures.
- Higher-resolution training for small-object detection.

Deployment Plan

- Cloud-based inference API.
- Web dashboard for terrain visualization.

Real-time Usage

Optimization for edge GPUs enabling real-time segmentation in autonomous systems.

Scalability

- Training on larger multi-terrain datasets.
 - Using larger Vision Transformer backbones for improved accuracy.
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